Hyperspectral Time-Series Target Detection Based on Spectral Perception and Spatial–Temporal Tensor Decomposition

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***Abstract*— The detection of camouflaged targets in the complex background is a hot topic of current research. The existing hyperspectral target detection algorithms do not take advantage of spatial information and rarely use temporal information. It is difficult to obtain the required targets, and the detection performance in hyperspectral sequences with complex background will be low. Therefore, a hyperspectral time-series target detection method based on spectral perception and spatial–temporal tensor (SPSTT) decomposition is proposed. First, a sparse target perception strategy based on spectral matching is proposed. To initially acquire the sparse targets, the matching results are adjusted by using the correlation mean of the prior spectrum, the pixel to be measured, and the four-neighborhood pixel spectra. The separation of target and background is enhanced by making full use of local spatial structure information through local topology graph representation of the pixel to be measured. Second, in order to obtain a more accurate rank and make full use of temporal continuity and spatial correlation, a spatial–temporal tensor (STT) model based on the gamma norm and $L_{2,1}$ norm is constructed. Furthermore, an excellent alternating direction method of multipliers (ADMM) is proposed to solve this model. Finally, spectral matching is fused with STT decomposition in order to reduce false alarms and retain more right targets. A 176-band Beijing Institute of Technology hyperspectral image sequence - I (BIT-HSIS-I) dataset is collected for the hyperspectral target detection task. It is found by testing on the collected dataset that the proposed SPSTT has superior performance over the state-of-the-art algorithms.**

***Index Terms*— Complex background, hyperspectral sequences, hyperspectral target detection, spatial–temporal tensor (STT) decomposition, spectral perception (SP).**

I. Introduction

In THE field of remote sensing, camouflaged target detection has attracted the attention of many researchers, and it has important applications in battlefield monitoring [1], [2]. However, traditional optical target detection techniques have the inherent limitation and cannot identify target with a similar appearance. For example, when the target is similar in shape and color to the background, the detection performance will be degraded. But, in some scenes, the texture signals of the background and the target are different [3], [4], [5], which can be used to extract the desired targets. The hyperspectral image (HSI) covers both spatial information and spectral information of ground objects. Spectral information of tens or even hundreds of bands reflects the subtle differences between different materials. These information can be employed to improve the recognition ability between target and background [6], [7], [8]. However, in complex background environments, HSI usually contains strong noise and texture-like objects, which are easy to cause false alarms [9]. Therefore, under the interference of complex ground background, how to reduce false alarms and improve detection performance have received more and more attention [10], [11], [12].

According to whether the prior spectral information is used, hyperspectral detection includes unsupervised anomaly detection and supervised target detection [13], [14], [15]. Anomaly detection refers to finding pixels that are discriminated from the global or local background pixels through pattern recognition or statistical methods without a clear target to be detected [16], [17], [18]. Since anomaly detection cannot find the specific targets of interest from the background, this article mainly studies hyperspectral target detection based on prior spectrum to detect specific camouflaged targets in complex backgrounds.

The existing hyperspectral target detection methods are mainly classified into three categories according to the degree of information utilization: spectral detection, spatial–spectral detection, and combined spatial–temporal spectral information detection [19], [20], [21]. At the beginning of the period, there are many target detection algorithms for spectral information [22]. Generalized likelihood ratio test (GLRT) algorithm was an important method to solve the hypothesis testing problem [23]. An adaptive cosine estimation (ACE) algorithm was proposed based on GLRT, which was based on an unstructured background model. When there were a few target pixels in the whole image, the influence on the detection result was weaker. As the number of target pixels in the entire image increased, the detection results of the algorithm would decrease significantly [24]. Farrand et al. developed the constrained energy minimization (CEM) algorithm. The main mean of the algorithm was to choose the signal in a specific direction, suppress the signal interference in other directions, and locate target pixels of needing from the image. This algorithm had a very significant effect on small target extraction, but the detection effect on large target was not

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good [25]. On the basis of this algorithm, Chang et al. realized a similar linear constraint-based minimum variance filtering algorithm to detect multidesired targets through a constraint vector. Matched filter (MF) algorithm was a constant false alarm probability detection algorithm on the basis of GLRT, which had a better detection effect in a simple background. However, for mixed-pixel targets, it was necessary to know the ratio of the target signal [26]. Based on the MF algorithm, an adaptive MF (AMF) algorithm was proposed, and its performance for the complex background was still poor [27]. Chang designed an orthogonal subspace projection (OSP) algorithm with the prior information of the target and background. The algorithm suppressed the background and highlighted the signal by projecting the pixel vectors onto an orthogonal subspace of the background subspace [28]. Heinz [29] put forward a detection method on the basis of constrained least-squares method according to the linear unmixing analysis method of pixels. Gu et al. focused on the kernel theory algorithm to extract nonlinear features in image through kernel principal component transformation. He used this method to reduce the dimensionality of HSI and then detect the target [30]. Some of these methods were based on linear detection, and some were based on nonlinear detection. Although the detection efficiency of these methods was high, they did not consider spatial–temporal information and have low detection performance in complex background datasets.

In recent years, due to the improvement of hyperspectral spatial resolution, some target detection methods with combined spatial and spectral information emerge [31]. Capobianco and Camps-Valls [32] constructed a kernel-OSP (KOSP) algorithm based on contextual information of combined kernel functions, which applied the pixel spatial neighborhood information to target detection. Zhao et al. released a spatial–spectral algorithm by background joint sparse representation (BJSR). Then, low-dimensional subspace manifolds of HSI were obtained by using a linear local tangent space alignment (LLTSA) algorithm for target detection [33]. Sun et al. proposed a tree-structured encoding method for hyperspectral target detection (TDTSE). It formed a binary tree based on the features of the data itself to segment the image and increase the separability of the target from the background [34]. Yang et al. developed an algorithm based on sparse-spatial CEM (SSCEM) for hyperspectral target detection, which extended the CEM method by comprehensively considering the spatial correlation and sparsity of the targets. This algorithm added total variation to the CEM algorithm and used the L_1 regularization part to make the output result sparse [35]. Yan et al. proposed a hyperspectral super-resolution for target enhancement. It involved mapping a specific spectral library, fusing the spectral library spectral information with the multi-spectral image (MSI) spatial information, and obtaining adaptive sparse coefficients [36]. Guo et al. proposed a target detection method based on antinoise hierarchical mutual-incoherence-induced discriminative (AHMID). It improved the separation of background and anomaly by maintaining the inherent structural information between them [37].

Among the methods that synthesize spatial–temporal spectral information, some researchers proposed sequence-based hyperspectral detection algorithms. Sequence detection suppresses background inhomogeneity and stationary unwanted targets, providing better detection performance under complex background conditions [38], [39], [40]. A sequence detection method mainly contains some low-rank sparse decomposition methods and tensor methods. In recent years, many scholars applied sequence detection to hyperspectral anomaly detection. Robust principal component analysis (RPCA) was proposed for moving target detection [41], [42], which transformed the target finding problem into the low-rank matrix recovery problem. Xu et al. [43] proposed gas plume detection by low-rank sparse as well as total variation (LRSTV). The method assumed that the background was stationary, and the targets were mobile, and the background and target were separated by the low-rank sparse method. In addition, the total variational regularization item had been employed for the constrained minimization problem by exploiting that the targets were continuous in time and space. Li et al. put forward a constrained sparse representation spatial–temporal (CSRST) anomaly detection method. The method consisted of spatial and temporal detectors. The final detection result was obtained by fusing the first two results. However, these algorithms using spatial–temporal information focused on anomaly detection and could not extract the specific target of interest. In addition, these algorithms did not utilize the overall structural information of the spatial–temporal data and gave little consideration to the fusion of spatial–temporal spectral information. They had low extraction ability for the interested motion targets and caused more false alarms in the detection results [44]. Considering that higher order tensors can more realistically represent the intrinsic structure of image [45], [46], a hyperspectral time-series target detection method based on spectral perception and spatial–temporal tensor (SPSTT) decomposition is proposed. This method considers the continuity of the target and the correlation of the background in different frames and makes full use of the spatial–temporal spectral information. The main contributions of the proposed SPSTT are summarized as follows.

1) A fusion model of SPSTT decomposition is constructed. The method not only utilizes prior spectral information, but also makes full use of spatial–temporal overall structural information.
2) In a spectral perception (SP) method, the matched filtering and local correlation fusion strategy are proposed to acquire the targets of interest. The topological graph representation is adapted to improve the distinction between target and background.
3) The spatial–temporal tensor (STT) is constructed by modeling the low-rank and sparse parts by using gamma norm and $L_{2,1}$ norm. In addition, we constructed the dataset Beijing Institute of Technology hyperspectral image sequence - I (BIT-HSIS-I), which was collected on the campus of Beijing Institute of Technology (BIT), Beijing, China, using the hyperspectral sensor GaiaSkymini2 from Dualix Spectral Imaging.

The remaining parts of this manuscript are summarized as follows. Section II details the proposed SPSTT. It includes sparse target perception based on spectral matching, STT

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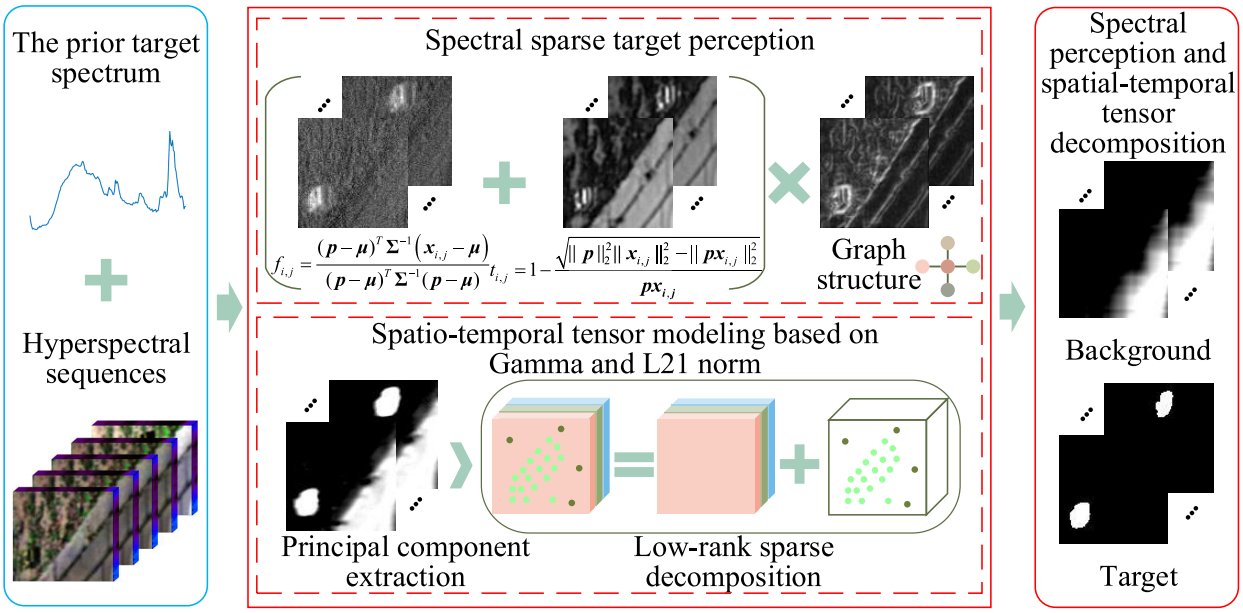


Fig. 1. Diagram of the proposed SPSTT for hyperspectral target detection, including spectral sparse target perception and STT modeling. SP consists of computing the correlation mean of the prior spectrum, the pixel to be measured, and the four-neighborhood pixel spectra, and adjusting it with the local topological graph representation. The STT model is constructed by modeling the low-rank and sparse parts by using the gamma norm and $L_{2,1}$ norm, respectively. The low-rank sparse decomposition of the model is implemented by ADMM.

decomposition based on gamma norm and $L_{2,1}$ norm, fusion based on SPSTT decomposition, and computational complexity analysis. In Section III, hyperspectral datasets are displayed, evaluation metrics are introduced, and detection performances are presented. In Section IV, the conclusion is presented.

II. Proposed Detection Methodology

The detection flow diagram of the proposed SPSTT is illustrated in Fig. 1. First, a spectral-based sparse target perception method is proposed. A weighted matching detection method based on correlation mean of the prior spectrum, the pixel to be measured, and the four-neighborhood pixel spectra is designed. The matching results are multiplied with the results of local topological graph representation. Next, the first principal component of each HSI is extracted with principal component analysis (PCA), and the STT cubes are constructed by modeling the low-rank and sparse parts with the gamma norm and $L_{2,1}$ norm, respectively. The target detection is equated to the low-rank and sparse decomposition (LRSD) solving problem. A better performing alternating direction method of multipliers (ADMM) is utilized to implement the solution to the model. Finally, the SP results are fused with the STT decomposition to obtain sparse targets.

A. Spectral Sparse Target Perception

Matched filtering can automatically find parts of target spectra that are similar to the prior spectrum in HSI. The matched filtering formula is stated

$$f_{i,j} = \frac{(\boldsymbol{p}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}\left(\boldsymbol{x}_{i,j}-\boldsymbol{\mu}\right)}{(\boldsymbol{p}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{p}-\boldsymbol{\mu})} \tag{1}$$

where i and j represent the coordinates of the pixel to be measured, $\boldsymbol{p}$ is the prior target spectrum, $\boldsymbol{\mu}$ is the mean of all tested pixels, $\boldsymbol{x}_{i,j}$ is the pixel to be measured, and $\boldsymbol{\Sigma}^{-1}$ is a covariance matrix of all tested pixels. To improve the accuracy and robustness of target perception, the tangents mean of the prior spectrum and the four-neighbor spectra are used to adjust the matching results. The formula for calculating the tangent

$$t_{i,j} = 1 - \frac{\sqrt{\|\boldsymbol{p}\|_2^2 \|\boldsymbol{x}_{i,j}\|_2^2 - \|\boldsymbol{p}^T \boldsymbol{x}_{i,j}\|_2^2}}{\boldsymbol{p}^T \boldsymbol{x}_{i,j}}. \tag{2}$$

The mean value of the tangents for the prior spectrum and the four-neighbor spectra is

$$\bar{t}_{i,j} = \frac{t(i-1,j)+t(i+1,j)+t(i,j)+t(i,j-1)+t(i,j+1)}{5}. \tag{3}$$

The matched filtering adjusted by tangents mean of the four-neighbor spectrum is

$$r_{i,j} = f_{i,j} + \bar{t}_{i,j}. \tag{4}$$

The detection results for all pixels are expressed as follows:

$$\boldsymbol{R} = \begin{bmatrix} r_{1,1} & \cdots & r_{1,n} \\ \vdots & \ddots & \vdots \\ r_{m,1} & \cdots & r_{m,n} \end{bmatrix} \tag{5}$$

where m and n are the number of rows and columns of the HSI, respectively. Graph topological information is able to represent the spatial structure relationship of an image by calculating the similarity between different pixels. The local graph representation is simple and efficient and has been applied in many fields, such as anomaly detection, unmixing, and band selection of HSI. In order to make full use of the local spatial information and further improve the separation between the target and the background, the matching results are adjusted by using the graph representation of the pixels to

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be measured and the pixels in their four neighborhoods. The local graph matrix is given by

$$T_{h,l} = \begin{bmatrix} \bar{t}_{1,1} & \cdots & \bar{t}_{1,5} \\ \vdots & \bar{t}_{h,l} & \vdots \\ \bar{t}_{5,1} & \cdots & \bar{t}_{5,5} \end{bmatrix} \tag{6}$$

where both h and l are from 1 to 5, $T_{h,l}$ denotes the similarity calculation between the hth pixel and the lth pixel in the local region, and $\bar{t}_{h,l}$ is given by

$$\bar{t}_{h,l} = \left\| \bar{t}_h - \bar{t}_l \right\| \tag{7}$$

where $\bar{t}_h$ denotes the root mean square of the value of the hth pixel. The local structure of the pixel $x(i, j)$ to be measured is expressed with the degree of that pixel node, which is calculated as follows:

$$\bar{r}_{i,j} = \sum_{l=1}^{5} T_{h,l} \tag{8}$$

where h is the pixel to be measured whose coordinates are (i, j). The local representation $\bar{r}_{i,j}$ of all pixels forms the matrix $\bar{\boldsymbol{R}}$. Combining the matching strategy with the graph structure representation strategy, the spectral sparse target perception result can be obtained with $\boldsymbol{S} = \boldsymbol{R}\bar{\boldsymbol{R}}$. The multiframe-based SP result tensor is

$$\boldsymbol{\mathcal{M}}(:, :, \nu) = \boldsymbol{S} \tag{9}$$

where $\boldsymbol{\mathcal{M}}(:, :, \nu)$ denotes the SP tensor at frame ν. The solution method of the proposed SP is summarized in Algorithm 1.

Algorithm 1 Solution Method of the Proposed SP Algorithm

Input: X: three-dimensional hyperspectral image; $\boldsymbol{p}$: prior target spectral; W: the number of pixels

Output: Single frame spectral perception result $\boldsymbol{S}$

1: **Initialization:** $\boldsymbol{x}_{i,j} = \boldsymbol{x}_k$, $W = mn$
2: **for** $k = 1 \ldots W$ **do**
3: Compute $f_{i,j}$ by Eq. (1);
4: Compute $\bar{t}_{i,j}$ by Eq. (3);
5: Compute $r_{i,j}$ by Eq. (4);
6: Compute $\bar{r}_{i,j}$ by Eq. (8);
7: Compute $\boldsymbol{\mathcal{M}}$ by Eq. (9);
8: $k \leftarrow k + 1$;
9: **end for**

Fig. 2 shows the 3-D detection results for matching algorithm and its weighting by local topological graph representation on sparse grass and brick data. Fig. 2(a)–(e) shows the matched filtering results adjusted by using the average of the tangent values of the prior spectrum, the pixel to be measured, and the four-neighborhood pixel spectra. Fig. 2(f)–(j) shows the results after weighting by the local topological graph representation. The results show that the target is more salient, and the separation of the target from the background is improved after weighting the local spatial structure information. The results provide prior constraints on the sparse target pixels obtained by the subsequent low-rank sparse decomposition algorithm, which can adjust the target detection probability.

B. STT Decomposition Based on Gamma and $L_{2,1}$ Norms

For motion target detection, there are existing algorithms based on LRSD models that exploit the existence of temporal continuity and spatial sparsity of targets across frames [47], [48]. The LRSD models are as follows:

$$\boldsymbol{X} = \boldsymbol{B} + \boldsymbol{T} \tag{10}$$

where $\boldsymbol{X}$ ($\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_W]$, $\boldsymbol{x}_W \in \mathbb{R}^{L \times W}$) is the raw HSI, $\boldsymbol{B}$ is the background, and $\boldsymbol{T}$ is the target. L and W are the numbers of frequency bands and of target pixels, respectively.

In traditional sequential target detection algorithms, each frame is reconstructed as a column vector of the matrix. These methods ignore structural information about the image itself. Therefore, a tensor-based model is proposed for hyperspectral sequential target detection. The model is constructed as follows:

$$\boldsymbol{\mathcal{X}} = \boldsymbol{\mathcal{B}} + \boldsymbol{\mathcal{T}} \tag{11}$$

where $\boldsymbol{\mathcal{X}}$, $\boldsymbol{\mathcal{B}}$, $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{Z_1 \times Z_2 \times Z_3}$, $\boldsymbol{\mathcal{X}}$ represents the original image tensor, $\boldsymbol{\mathcal{B}}$ represents the background tensor, and $\boldsymbol{\mathcal{T}}$ represents the target tensor. Here, Z_1, Z_2, and Z_3 denote the numbers of rows, columns, and frames, respectively. As shown in Fig. 3, the singular value of the STT along the model-1, model-2, and model-3 expansions is plotted. The low rankness of these modes along the subspace can be easily shown from the fast convergence properties of these curves. Therefore, the hyperspectral target detection problem is transformed into the following optimization problem:

$$\begin{aligned} &\min_{\boldsymbol{\mathcal{B}}, \boldsymbol{\mathcal{T}}} \operatorname{rank}(\boldsymbol{\mathcal{B}}) + \beta \|\boldsymbol{\mathcal{T}}\|_0 \\ &\text{s.t. } \boldsymbol{\mathcal{X}} = \boldsymbol{\mathcal{B}} + \boldsymbol{\mathcal{T}} \end{aligned} \tag{12}$$

where β is a positive parameter that controls sparsity and $||\ ||_0$ denotes L_0 norm. For (12), the solution is an NP-hard problem. Considering the sparse distribution of hyperspectral targets in HSI, if the background components are removed, the data are dense in the temporal dimension and sparse in the spatial dimension. Therefore, the target component has a group sparse property. So, the L_0 norm is considered to be replaced by $L_{2,1}$ norm. The above formula is rewritten as follows:

$$\begin{aligned} &\min_{\boldsymbol{\mathcal{B}}, \boldsymbol{\mathcal{T}}} \operatorname{rank}(\boldsymbol{\mathcal{B}}) + \beta \|\boldsymbol{\mathcal{T}}\|_{2,1} \\ &\text{s.t. } \boldsymbol{\mathcal{X}} = \boldsymbol{\mathcal{B}} + \boldsymbol{\mathcal{T}} \end{aligned} \tag{13}$$

where $L_{2,1}$ norm is the L_2 norm for the column (temporal domain) and the L_1 norm for the row (spatial domain). It is defined as $\|\boldsymbol{\mathcal{T}}\|_{2,1} = \sum_{j=1}^{mn} (\sum_{p=1}^{t} t_{p,q}^2)^{1/2}$. Here, $t_{p,q}$ is the pth row and qth column element in the mode-3 expansion of $\boldsymbol{\mathcal{T}}$.

The researchers had done some research on the solution of the low-rank part. Zhang et al. [49] constructed a tensor kernel norm (TNN) penalty algorithm. Liu et al. [50] deduced the equivalence relationship between the low-rank tensor and the nuclear norm of its core tensor and then used the nuclear norm to calculate the rank of the whole tensor. Kong et al. [51] tried to use the Schatten-p (SHP) norm to ensure accurate signal recovery. However, these methods failed to accurately approximate the value of rank, and Cai et al. [52] proposed a tensor low-rank approximation method based on the gamma norm, which was shown to have significant performance on

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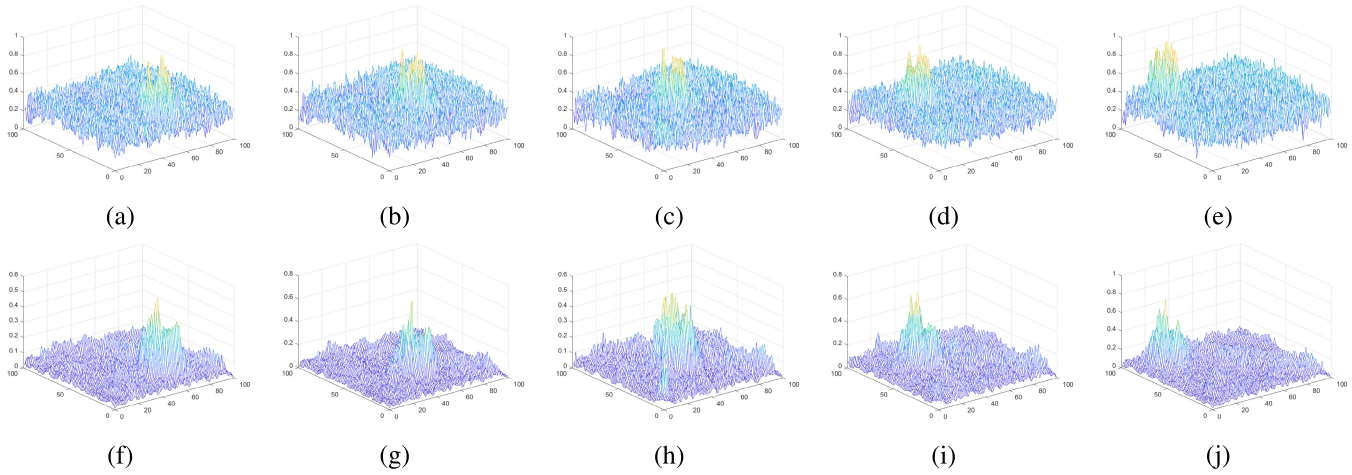


Fig. 2. 3-D detection results for matching algorithm and its weighting by local topological graph representation on sparse grass and brick data. (a)–(e) Results for matching detection. (f)–(j) Results for local topology graph representation algorithm-weighted matching detection.

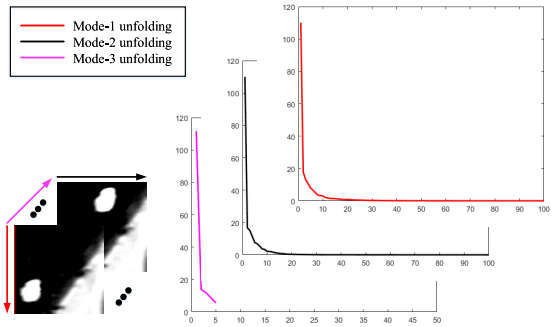


Fig. 3. Image of the singular value distribution of the STT unfolds along model-1, model-2, and model-3.

video background modeling. Therefore, this method is utilized for modeling the low-rank component, and the STT model is represented as follows:

$$\min_{\boldsymbol{\mathcal{B}},\boldsymbol{\mathcal{T}}} \|\boldsymbol{\mathcal{B}}\|_{\gamma} + \beta\|\boldsymbol{\mathcal{T}}\|_{2,1} \quad \text{s.t. } \boldsymbol{\mathcal{X}} = \boldsymbol{\mathcal{B}} + \boldsymbol{\mathcal{T}} \tag{14}$$

where $\|\boldsymbol{\mathcal{B}}\|_{\gamma}$ is

$$\|\boldsymbol{\mathcal{B}}\|_{\gamma} = \frac{1}{Z_3}\sum_{z_3=1}^{Z_3}\left\|\hat{B}_{z_3}\right\|_{\gamma} = \frac{1}{Z_3}\sum_{z_3=1}^{Z_3}\left(\sum_{z=1}^{\min\{Z_1,Z_2\}}\frac{(1+\gamma)\hat{\Sigma}_{z,z,z_3}}{\gamma+\hat{\Sigma}_{z,z,z_3}}\right) \tag{15}$$

where $\hat{B}_{z_3}$ is the z_3th frontal slice of $\widehat{\boldsymbol{\mathcal{B}}} = \text{FFT}(\boldsymbol{\mathcal{B}}, [], 3) = [\widehat{B}_1 \cdots \widehat{B}_{Z_3}]$ and FFT denotes the fast Fourier transform.

We use ADMM to solve (14). Constructing its augmented Lagrangian formula, the formula can be expressed as follows:

$$\begin{aligned}\text{LF}(\boldsymbol{\mathcal{B}},\boldsymbol{\mathcal{T}},\boldsymbol{\mathcal{Q}},\mu) &= \|\boldsymbol{\mathcal{B}}\|_{\gamma} + \beta\|\boldsymbol{\mathcal{M}}\odot\boldsymbol{\mathcal{T}}\|_{2,1} + \langle\boldsymbol{\mathcal{Q}},\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{B}}-\boldsymbol{\mathcal{T}}\rangle + \frac{\mu}{2}\|\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{B}}-\boldsymbol{\mathcal{T}}\|_F^2 \\ &= \|\boldsymbol{\mathcal{B}}\|_{\gamma} + \beta\|\boldsymbol{\mathcal{M}}\odot\boldsymbol{\mathcal{T}}\|_{2,1} + \frac{\mu}{2}\left\|\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{B}}-\boldsymbol{\mathcal{T}}+\frac{\boldsymbol{\mathcal{Q}}}{\mu}\right\|_F^2 - \frac{1}{2\mu}\|\boldsymbol{\mathcal{Q}}\|_F^2\end{aligned} \tag{16}$$

where LF denotes the augmented Lagrangian function, $\boldsymbol{\mathcal{Q}}$ denotes the Lagrangian multiplier, and β and μ denote the positive penalty factors. The ADMM framework is used to update $\boldsymbol{\mathcal{B}}$, $\boldsymbol{\mathcal{T}}$, $\boldsymbol{\mathcal{Q}}$, and μ, respectively.

1) After fixing the other parameters, $\boldsymbol{\mathcal{B}}$ is updated by computing the following equation:

$$\begin{aligned}\boldsymbol{\mathcal{B}}_{k+1} &= \arg\min_{\boldsymbol{\mathcal{B}}} \|\boldsymbol{\mathcal{B}}\|_{\gamma} + \frac{\mu}{2}\left\|\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{B}}-\boldsymbol{\mathcal{T}}_k+\frac{\boldsymbol{\mathcal{Q}}_k}{\mu_k}\right\|_F^2 \\ &= \arg\min_{\boldsymbol{\mathcal{B}}} \frac{2}{\mu}\|\boldsymbol{\mathcal{B}}\|_{\gamma} + \left\|\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{B}}-\boldsymbol{\mathcal{T}}_k+\frac{\boldsymbol{\mathcal{Q}}_k}{\mu_k}\right\|_F^2.\end{aligned} \tag{17}$$

The solution of this equation is

$$\boldsymbol{\mathcal{B}}_{k+1} = \mathcal{U}_{\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{T}}_k+\frac{\mathcal{O}_k}{\mu_k}} * \Omega_{\boldsymbol{\mathcal{B}}} * \mathcal{V}_{\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{T}}_k+\frac{\mathcal{O}_k}{\mu_k}} \tag{18}$$

where $\mathcal{U}_{\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{T}}_k+(\mathcal{O}_k/\mu_k)}$ and $\mathcal{V}_{\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{T}}_k+(\mathcal{O}_k/\mu_k)}$ are obtained by the singular value decomposition (SVD) of $\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{T}}_k+(\mathcal{O}_k/\mu_k)$, $\Omega_{\boldsymbol{\mathcal{B}}} \in \mathbb{R}^{Z_1\times Z_2\times Z_3}$ is the diagonal tensor, $1 \le z \le \min(Z_1, Z_2)$, $1 \le z_3 \le Z_3$, and $\hat{\Sigma}$ and $\Omega_{\boldsymbol{\mathcal{B}}}$ satisfy the following equation:

$$\begin{aligned}\hat{\Sigma} &= \left[\hat{\Sigma}_{z,z,z_3}\right] = \text{FFT}(\Sigma, [], 3) \\ \hat{\Omega}_{\boldsymbol{\mathcal{B}}} &= \left[\hat{\Omega}_{z,z,z_3}\right] = \text{FFT}(\Omega, [], 3)\end{aligned} \tag{19}$$

where $\text{FFT}(\Omega, [], 3)$ denotes applying FFT on Σ or Ω along the third mode. The $\hat{\Omega}_{z,z,z_3}$ is calculated as follows:

$$\hat{\Omega}_{z,z,z_3}^{(k+1)} = \left(\hat{\Sigma}_{z,z,z_3} - \tau\frac{\gamma(1+\gamma)}{\left(\gamma+\hat{\Omega}_{z,z,z_3}^{(k)}\right)^2}\right)_+ \tag{20}$$

where $\hat{\Omega}_{z,z,z_3}$ is the limit point of fix point iteration and the meaning of the $+$ at the end of (20) is that the right limit of the formula tends to 0.

2) After fixing the other parameters, $\boldsymbol{\mathcal{T}}$ is updated by calculating the following equation:

$$\begin{aligned}\boldsymbol{\mathcal{T}}_{k+1} &= \beta\|\boldsymbol{\mathcal{M}}\odot\boldsymbol{\mathcal{T}}\|_{2,1} + \frac{\mu}{2}\left\|\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{B}}_{k+1}-\boldsymbol{\mathcal{T}}+\frac{\boldsymbol{\mathcal{Q}}_k}{\mu_k}\right\|_F^2 \\ &= \frac{\beta}{\mu}\boldsymbol{\mathcal{M}}\|\boldsymbol{\mathcal{T}}\|_{2,1} + \frac{1}{2}\left\|\boldsymbol{\mathcal{X}}-\boldsymbol{\mathcal{B}}_{k+1}-\boldsymbol{\mathcal{T}}+\frac{\boldsymbol{\mathcal{Q}}_k}{\mu_k}\right\|_F^2.\end{aligned} \tag{21}$$

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The solution of the above equation is

$$\mathcal{T}_{k+1} = \phi_{\beta\mathcal{M}/\mu}\left(\mathcal{X} - \mathcal{B}_{k+1} + \frac{Q_k}{u_k}\right) \tag{22}$$

where m_q denotes the qth column of the model-3 expansion of $\mathcal{X} - \mathcal{B}_{k+1} + (Q_k/u_k)$. The optimal solution t_q^* of the jth column of the model-3 expansion of $\mathcal{T}_{k+1}$ is

$$t_q^* = \begin{cases} \dfrac{\|m_q\| - \beta\mathcal{M}/\gamma}{\|m_q\|} m_q, & \text{if } \|m_q\| < \|\beta\mathcal{M}/\gamma\| \\ 0, & \text{otherwise.} \end{cases} \tag{23}$$

3) The Lagrangian multiplier $\mathcal{Q}$ is updated as follows. The equation $< \mathcal{Q}, \mathcal{X} - \mathcal{B}_{k+1} - \mathcal{T}_{k+1} >$ is derived. $\mathcal{Q}_{k+1}$ is solved as follows:

$$\mathcal{Q}_{k+1} = \mathcal{Q}_k + \mu_k(\mathcal{X} - \mathcal{B}_{k+1} - \mathcal{T}_{k+1}). \tag{24}$$

4) The parameter μ is updated as follows:

$$\mu_{k+1} = \mu_k \rho. \tag{25}$$

The solution method of the proposed STT is summarized in Algorithm 2.

Algorithm 2 Solution Method of the Proposed STT Algorithm

Input: The original image tensor $\mathcal{X}$, the spectral perception prior tensor $\mathcal{M}$, the parameters λ, μ^0, γ, ρ and $k = 0$

Output: Target detection result $\mathcal{T}$

```
while not convergence do
    Update B_{k+1} by Eq. (18);
    Update T_{k+1} by Eq. (22);
    Update Q_{k+1} by Eq. (24);
    Update μ_{k+1} by Eq. (25);
    i ← i + 1;
end while
```

The iteration stops when the reconstructed residuals of the proposed STT algorithm satisfies the following equation:

$$\|\mathcal{B}_{k+1} + \mathcal{T}_{k+1} - \mathcal{X}\| = c(\mu^{(0)}\rho^{-k/2}) \tag{26}$$

where c is a constant, μ^0 is the initial value of 10^{-4}, and ρ is 1.1. The iteration is stopped when the difference between the computational tensor of two adjacent iterations is less than the specified appropriate threshold $\Im$ (the value is 10^{-8}), and the number of computations required for the iteration is

$$T = O\left(\log \rho^{(\mu^{(0)}/\Im)}\right). \tag{27}$$

Fig. 4 shows the reconstructed residuals and the number of iterations of the proposed algorithm STT solved on SGB data. From Fig. 4, it can be found that when the number of iterations is 78, the reconstruction residuals satisfy the set threshold, the iterations stop, and the proposed STT converges.

C. Fusion of SPSTT Decomposition

The SP can be used to obtain targets of specific desired materials with a prior spectra, and the STT model can be used to obtain moving targets by using spatial–temporal correlation. In order to make full use of spatial–temporal spectral information to reduce false alarms and improve target detection probability, the two methods are combined. The formulation of the proposed fusion method can be expressed as follows:

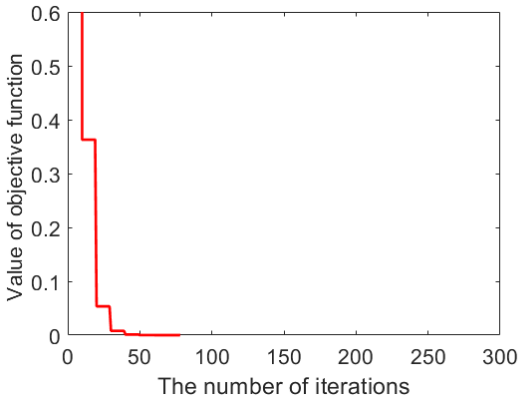


Fig. 4. Reconstructed residuals of the proposed STT for the iterative solution on SGB data.

$$\mathcal{R} = \mathcal{M} + \mathcal{T} \tag{28}$$

where $\mathcal{M}$ is the spectrum perception prior tensor.

III. Experiments and Analysis

This section includes hyperspectral datasets, performance evaluation criteria, parameter analysis, and detection results of the proposed SPSTT. The comparison algorithms used in this experiment mainly include CEM [25], MF [26], ACE [24], TDTSE [34], and AHMID [37]; and sequence-based detection methods include LRSTV [43], CSRST [44], TNN [49], and SHP [51].

A. Hyperspectral Datasets

A new BIT-HSIS-I dataset is collected for the detection experiments. The prior target spectra are acquired from close-range HSI of cars. Here, the mean of the spectrum of the center pixel and its four-neighbor pixels of a camouflaged car are used as the prior spectrum. This dataset was collected on the campus of BIT. The sensor used in this experiment is the Dualix hyperspectral GaiaSkymini2. The spectral range is 400–1000 nm, and its resolution is 3.5 nm. When the distance is 500 m, the spatial resolution is 0.14 m. The number of spectral bands is 176. Each band contains 150 × 150 pixels. The detection effect is excellent when the number of frames is five frames. The target being detected is a camouflaged car model. The backgrounds are sparse grass and brick, bare soil and concrete walls, shadow grass and water snow, and dry grass and shrub, respectively. The detailed introductions of the BIT-HSIS-I are listed in Table I.

B. Evaluation Metrics and Analysis

In order to better evaluate the detection ability of the algorithms in this article, some evaluation metrics are employed. These evaluation indicators mainly include 2-D and 3-D detection result maps, target background box plot of all sequences of detection results, receiver operating characteristic (ROC)

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TABLE I
HYPERSPECTRAL TARGET DETECTION EXPERIMENTAL DATASETS COLLECTED BY THE BIT GROUP

Image	Collection date	Spectral resolution	Size of space	Sensor	Background description
Sequence 1	23/10/2021	3.5nm ± 0.5nm	100×100	GaiaSkymini2	Sparse grass and bricks
Sequence 2	08/11/2021	3.5nm ± 0.5nm	150×150	GaiaSkymini2	Shadow grass and water snow
Sequence 3	20/10/2021	3.5nm ± 0.5nm	150×150	GaiaSkymini2	Bare soil and cement walls
Sequence 4	31/10/2021	3.5nm ± 0.5nm	150×150	GaiaSkymini2	Dry grass and shrub

curve, detection time, and area under the curve (AUC). The 2-D and 3-D detection result maps are gray-scale and mesh maps, respectively. They directly reflect the location of the target. The quality of the detection algorithm is qualitatively determined by the number of true bright pixels.

The separation of target and background is represented by the box plot [53]. In all box plots, median values are represented by the centerline. The 75th and 25th of the data maximum are represented by the upper and lower edges of the box. Whisker refers to the extreme value of the detection result. The red box represents the target range, and the green box represents the background range.

The point on the ROC curve represents the probability of detection (PD) at the different probabilities of false detection (PFs) [54], [55]. The closer the ROC is to the top left, the larger the PD and the smaller the PF, indicating that the detection performance of the related algorithm is better. PD is obtained by the ratio of the number of the correctly detected target pixels to the total number of actually owned target pixels. PF is obtained by the ratio of the number of the falsely detected target pixels to the number of all pixels. Above PF and PD are calculated by

$$\mathrm{PD} = \frac{N_t}{N_a} \tag{29}$$

$$\mathrm{PF} = \frac{N_f}{N_l} \tag{30}$$

where N_t, N_a, N_f, and N_l represent the number of correctly detected target pixels, the number of real target pixels, the number of wrongly detected pixels, and the number of all pixels of the image, respectively.

After obtaining the required arrays corresponding to PD and PF, the value of AUC is deduced [56]. The AUC refers to the area surrounded by ROC and the X-axis. The h indicates the number of all thresholds. The larger the AUC is, the better the detection effect. AUC is calculated by

$$\mathrm{AUC} = \frac{1}{2} \times \sum_{i}^{h} (\mathrm{PF}_i - \mathrm{PF}_{i-1}) \times (\mathrm{PD}_i + \mathrm{PD}_{i-1}). \tag{31}$$

The proposed algorithm involves some parameter selection. Fig. 5 analyzes the effect of different parameters λ, γ, and s on the proposed algorithm. In Fig. 5(a), $\lambda = l \cdot (\max(Z_1, Z_2)Z_3)^{1/2}$, where l is set to 1–5. The result is superior when l is 2. In Fig. 5(b), the value of $\gamma = g \cdot (\min(Z_1, Z_2))^{1/2}$, where g is set to 1–5, and the result is good when g is equal to 2. In Fig. 5(c), when the number of sequences s of HSIs is 5, the detection performance is excellent. Overall, these parameters have some effects on the results within these range. In practice, five adjacent frames are used to construct tensors, and smaller values of these parameters are chosen.

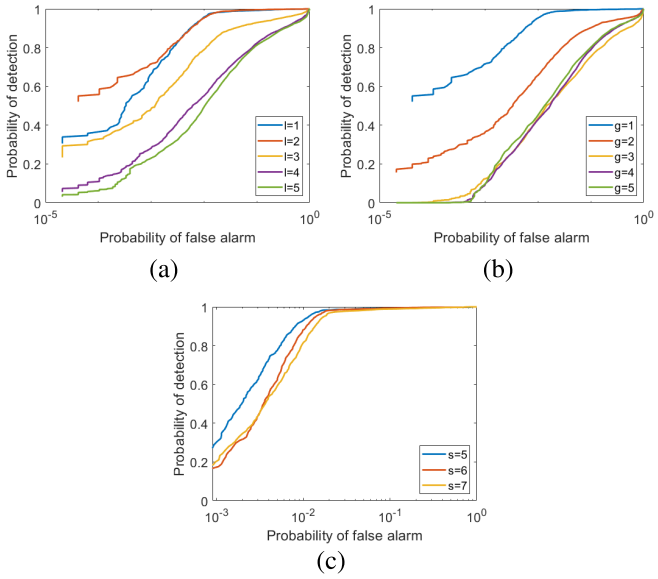


Fig. 5. ROC curves when the experimental parameters involved are chosen at different values. (a) Parameter λ. (c) Parameter γ. (c) Sequence s.

C. Detection Performance Comparison

Fig. 6 illustrates the ROC curves of the proposed SP, STT, and SPSTT. On the four datasets, the ROC curves of the proposed method SP, STT, and SPSTT are all close to the top left, indicating that the performance of all the proposed algorithms is good. Among the four imageries, the ROC curve of the SPSTT algorithm is closest to the top left, which shows the best performance of the spatial–temporal spectrum fusion algorithm.

Figs. 7(c)–(j) and 8(c)–(j) illustrate the 2-D and 3-D detection results by using different methods in four scenarios. To make it easier to compare the performance of these algorithms, they are all chosen to binarize the results at the PF of 0.001. The red rectangles are utilized to label the detection target pixels, and the blue ellipses are utilized to label the detected error pixels. The background corresponding to Fig. 7 is sparse grass and brick. In Fig. 7(c)–(f), the target pixels detected by CEM, MF, ACE, and TDTSE are sparse, and many wrong target pixels are detected. In Fig. 7(g)–(i), the target pixels detected by AHMID, LRSTV, and CSRST are relatively complete, but many wrong target pixels are detected. In Fig. 7(j), the proposed SPSTT not only can detect all targets, but also does not cause false alarms. The background corresponding

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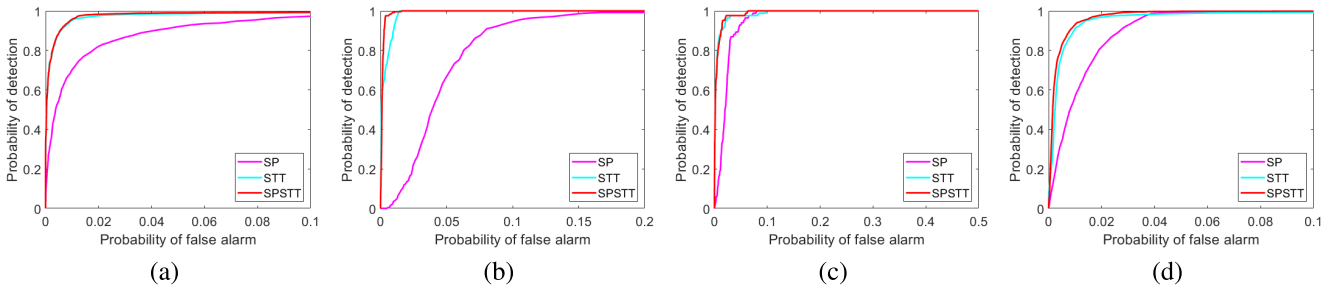


Fig. 6. ROC curves of the proposed SP, STT, and the SPSTT models on four datasets. (a) Sparse grass and brick. (b) Shadow grass and water snow. (c) Bare soil and cement wall. (d) Dry grass and shrub.

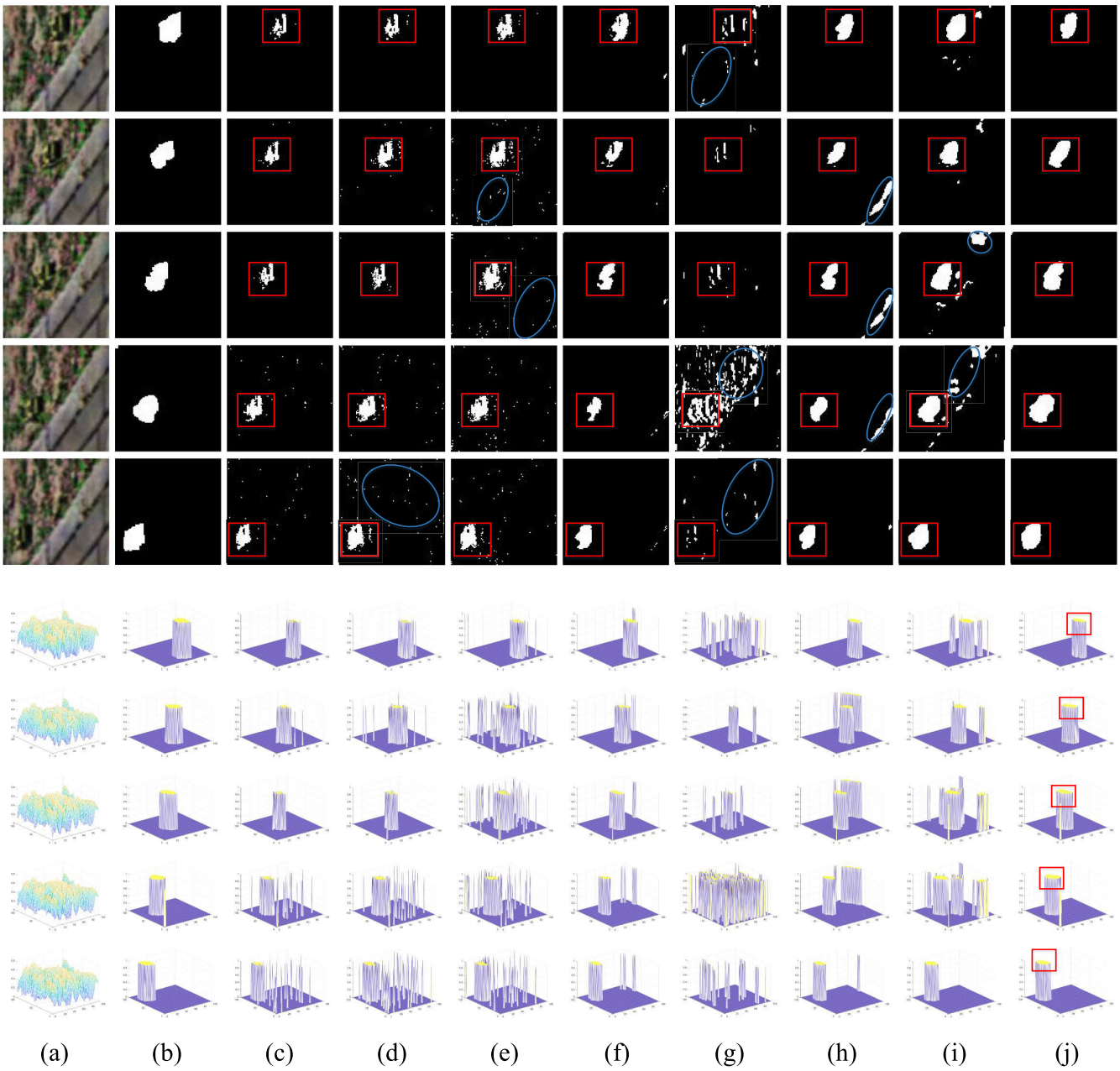


Fig. 7. 2-D and 3-D detection results of algorithms in sparse grass and brick background when PFs are all 0.001. (a) Pseudo-color image. (b) Ground-truth map. (c) CEM. (d) MF. (e) ACE. (f) TDTSE. (g) AHMID. (h) LRSTV. (i) CSRST. (j) SPSTT.

to rows 1 and 2 in Fig. 8 is shadow grass and water snow. In rows 1 and 2 of Fig. 8(e), (f), and (h), ACE, TDTSE, and LRSTV detect some target pixels and many wrong target pixels. In rows 1 and 2 of Fig. 8(c), (d), (g), and (i), CEM, MF, AHMID, and CSRST rarely detect the correct target pixels. In rows 1 and 2 of Fig. 8(j), the proposed SPSTT is

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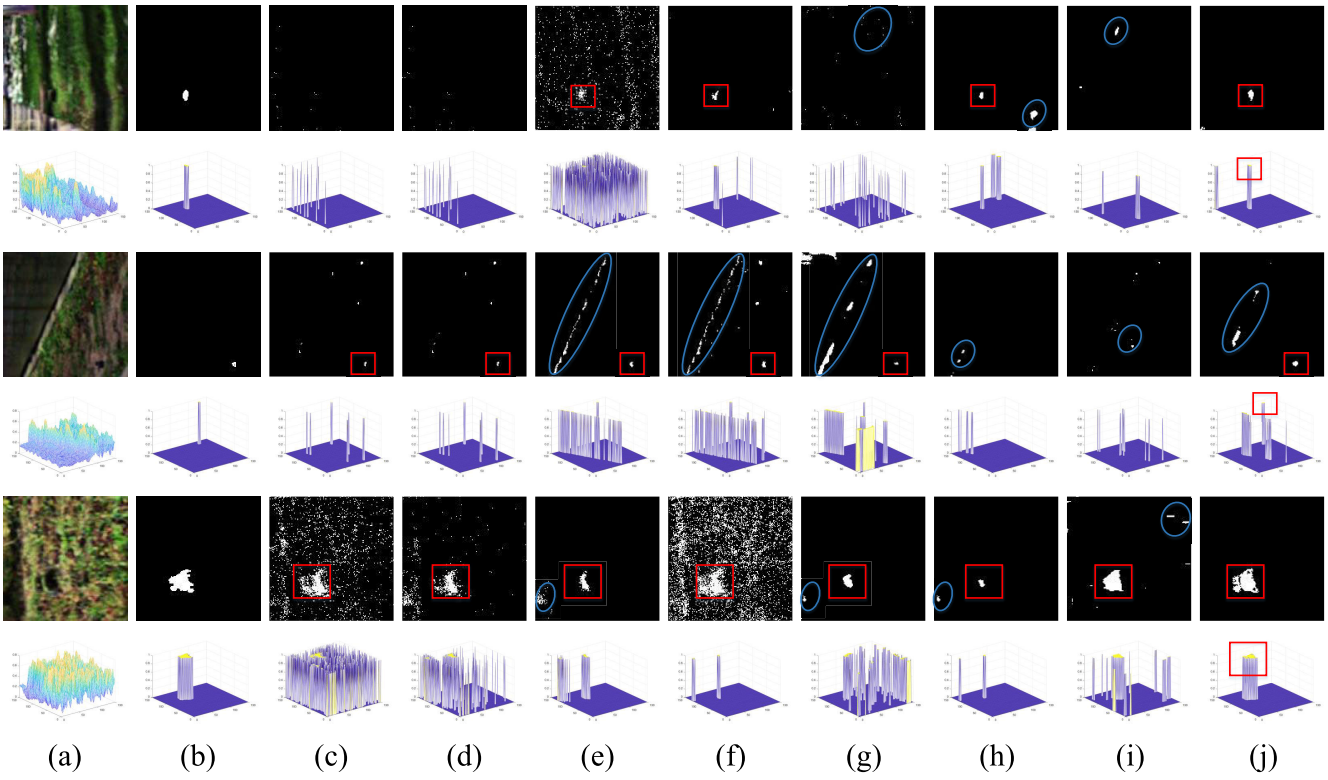


Fig. 8. 2-D and 3-D detection results of algorithms in shadow grass and water snow, bare soil and cement wall, and dry grass and shrub background when PFs are all 0.001. (a) Pseudo-color image. (b) Ground-truth map. (c) CEM. (d) MF. (e) ACE. (f) TDTSE. (g) AHMID. (h) LRSTV. (i) CSRST. (j) SPSTT.

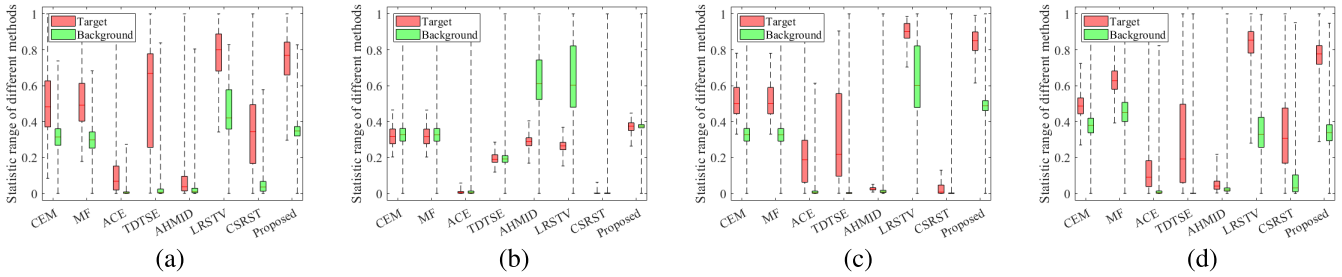


Fig. 9. Separation diagrams of methods in four scenes. (a) Sparse grass and brick. (b) Shadow grass and water snow. (c) Bare soil and cement wall. (d) Dry grass and shrub.

able to detect the whole target pixels with few false target pixels. Rows 3 and 4 of Fig. 8 are bare soil and cement wall background. In rows 3 and 4 of Fig. 8(c)–(g), CEM, MF, ACE, TDTSE, and AHMID detect few target pixels and many false target pixels. In rows 3 and 4 of Fig. 8(h) and (i), most of the pixels detected by LRSTV and CSRST are wrong. In rows 3 and 4 of Fig. 8(j), the proposed SPSTT is able to detect all objects with a few false alarms. Rows 5 and 6 of Fig. 8 are dry grass and shrub background. In rows 5 and 6 of Fig. 8(c), (d), (f), and (i), CEM, MF, TDTSE, and CSRST detect most of the target pixels and many false pixels. In rows 5 and 6 of Fig. 8(e), (g), and (h), ACE, AHMID, and LRSTV detect few wrong pixels and a few target pixels. In rows 5 and 6 of Fig. 8(j), the proposed SPSTT detects almost all targets with few false alarms. From the visual aspect, the proposed SPSTT has better detection performance. The proposed SPSTT fusion strategy can make full use of the spectral and spatial–temporal information to increase the difference between the background and the target.

In the separation diagram, the distance between the red box and the green box indicates the separation performance of the target and the background in the detection result. In Fig. 9(a) and (c), the spacing between red and green of single-frame detection methods (i.e., CEM, MF, ACE, TDTSE, and AHMID) is smaller. The sequence detection methods (i.e., LRSTV and CSRST) have larger red–green box spacing. The proposed SPSTT has the largest red–green box spacing. In Fig. 9(b), the red–green box spacing of SPSTT is smaller than that of AHMID and LRSTV, but its target range is larger than other algorithms. In Fig. 9(d), the red–green box spacing of the SPSTT algorithm is similar to LRSTV, and the red–green box spacing of the remaining algorithms is lower than the SPSTT algorithm. Therefore, the proposed SPSTT is easy to separate the required target from the background.

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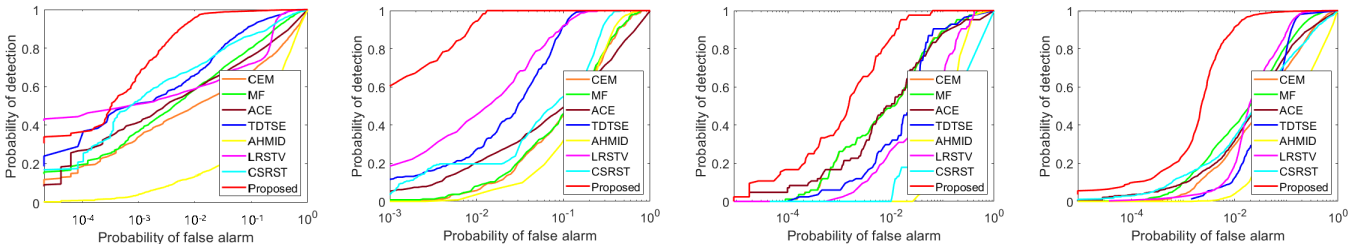


Fig. 10. ROC curves of methods in four scenes. (a) Sparse grass and brick. (b) Shadow grass and water snow. (c) Bare soil and cement wall. (d) Dry grass and shrub.

TABLE II
AUC VALUES OF DIFFERENT DETECTORS ON FOUR DATASETS

Image	CEM	MF	ACE	TDTSE	AHMID	LRSTV	CSRST	SP	STT	SPTNN	SPSHP	SPSTT
Sparse grass and brick	0.8417	0.9246	0.8835	0.9677	0.7221	0.9251	0.9460	0.9848	0.9944	0.9848	0.9853	0.9956
Shadow grass and water snow	0.8165	0.8235	0.7753	0.9599	0.8076	0.9691	0.8819	0.9530	0.9949	0.9530	0.9556	0.9985
Bare soil and cement wall	0.9577	0.9576	0.9357	0.9465	0.8290	0.9055	0.7090	0.9778	0.9912	0.9778	0.9853	0.9951
Dry grass and shrub	0.8939	0.9404	0.9051	0.9333	0.7447	0.9601	0.8887	0.9884	0.9930	0.9890	0.9880	0.9965

TABLE III
COMPUTATIONAL COMPLEXITY OF DIFFERENT DETECTORS ON FOUR DATASETS

Method	CEM	MF	ACE	TDTSE	AHMID	LRSTV	CSRST	SPTNN	SPSHP	SPSTT
Complexity	$O(NZ_3)$	$O(NZ_3)$	$O(NZ_3)$	$O(NZ_3)$	$O(\omega_1 {N_b}^2 NZ_3)$	$O(\omega_2 (NZ_3)^2)$	$O(\omega_3 NZ_3)$	$O(\omega_4 NZ_3)$	$O(\omega_5 NZ_3)$	$O(\omega_6 NZ_3)$

TABLE IV
EXECUTION TIME OF DIFFERENT DETECTORS ON FOUR DATASETS (UNIT: SECONDS)

Image	CEM	MF	ACE	TDTSE	AHMID	LRSTV	CSRST	SPTNN	SPSHP	SPSTT
Sparse grass and brick	0.2037	0.4812	0.7276	22.0980	8.9763	1126.0214	32.7720	16.1309	15.9738	17.0549
Shadow grass and water snow	0.6417	0.9458	1.4242	69.6179	31.5450	2414.2579	38.6115	35.0905	31.6496	34.3976
Bare soil and cement wall	0.3761	0.5741	0.8414	69.2912	35.6501	2245.3912	61.7917	40.4324	39.2175	39.6438
Dry grass and shrub	0.6626	1.0906	0.8554	68.7926	23.3583	2420.0021	42.0859	41.7071	53.9115	43.7870

Fig. 10 shows the ROC curves of the methods involved for all sequences on four different datasets. As shown in Fig. 10(a) (sparse grass and brick), when the PF approaches 0.1, the PD corresponding to the proposed SPSTT is about 0.9926. The PD*s* of other methods CEM, MF, ACE, TDTSE, AHMID, LRSTV, and CSRST are 0.6835, 0.8046, 0.7773, 0.9169, 0.3527, 0.7230, and 0.8691, respectively. When the PF is greater than 10^{-3}, the ROC curve of the proposed SPSTT is above other algorithms. From Fig. 10(b)–(d), when PF*s* ar 0.1, the PD*s* of the proposed SPSTT are all 1.0000. At this time, the maximum PD*s* of other mentioned algorithms are 0.8991, 0.8929, and 0.9211, respectively. When the PF is greater than 10^{-4}, the ROC curve of the proposed algorithm is closer to the top left of the coordinate axis. The ROC curves of above four datasets verify the effectiveness of the proposed SPSTT.

Table II lists the AUC values of the above comparison algorithms for all sequences. In the sparse grass and brick background, the AUC value of the proposed SPSTT is 0.9956. For the remaining baseline algorithms, TDTSE has the largest AUC value of 0.9677, and AHMID has the smallest AUC value of 0.7221. The AUC values of the proposed SPSTT for the remaining different scenarios with shadow grass and water snow, bare soil and cement wall, and dry grass and shrub are 0.9985, 0.9951, and 0.9965, respectively. The optimal AUC values of the other baseline algorithms in the remaining three scenarios are 0.9691, 0.9577, and 0.9601, respectively. The AUC values of the proposed SP, STT, and SPSTT are all exceeded the other baseline algorithms in all scenarios. The AUC of the proposed SPSTT method is also superior to the combination of SP with TNN and SHP (SPTNN and SPSHP).

The computational complexity is listed in Table III. N denotes the number of pixels, Z_3 is the number of sequences, and N_b is the number of background dictionaries. ω_1, ω_2, ω_3, ω_4, ω_5, and ω_6 are the number of iterations of the corresponding method. The computational complexity of LRSTV is the highest. The computational complexity of the proposed method is similar to the baseline method. The execution time is listed in Table IV. In four different scenarios for all sequences, the computational cost by the proposed SPSTT is 17.0549, 34.3976, 39.6438, and 43.7870 s, respectively. The main computational consumption of the proposed SPSTT comes from the local distance graph computation, and the iterative computation of the reconstructed residuals is less consumed. The computational cost by LRSTV and CSRST all exceeds SPSTT. The detection time required by LRSTV is the longest, which is about 2420 s. From the above analysis,

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the proposed SPSTT has a better performance when the time consumption is small. All detection algorithms are run in MATLAB on an Intel Core i7-8700 h CPU computer with 8 Gb of computer memory.

IV. Conclusion

In this article, a hyperspectral time-series target detection method based on SPSTT decomposition was proposed. First, sparse target pixels were obtained by using a spectral matching detection algorithm that could detect specific desired material camouflage targets. The matching results were adjusted with the correlation mean of the prior target, the pixel to be measured and its four neighboring pixels, and some sparse targets were initially screened. The local spatial information was fully utilized by using local topological graph representation to increase the separation of background and target. Second, hyperspectral sequences were used to construct an STT model based on the gamma norm and $L_{2,1}$ norm. The model not only made full use of the background correlation but also took full advantage of the temporal continuity of the target. Finally, the fusion of the SP prior and the STT model could detect more accurate target pixels. Experimental results on the new BIT-HSIS-I dataset showed that the proposed SPSTT had better detection performance than other comparative algorithms.

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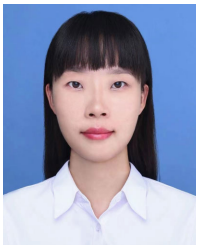
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